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MIL-STD-2045-14502-6A
27 July 1995

MILITARY STANDARD

INFORMATION TECHNOLOGY
DOD STANDARDIZED PROFILE
INTERNET TRANSPORT PROFILE
FOR DOD COMMUNICATIONS

PART 6: Combat Net Radio

AMSC N/A

AREA DCPS

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Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD). All previous versions of this document are hereby superseded.

Beneficial comments (recommendations, additions, deletions) and any pertinent data for improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: JEBBF
Squier Hall, Bldg 283
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This DSP is a functional DOD Data Communications Protocol Standard (DCPS) produced by the DCPS Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to DOD.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, is used to describe how DOD implements commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Profiles for Open Systems Internetworking Technologies (POSIT) will serve as the base for developing the 10000 series with DOD enhancements. POSIT has superseded the Government Open Systems Interconnection Profile (GOSIP). Unique military standards will be used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, is used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, is used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, is used to document interim standards. Interim standards document protocols and services needed by DOD until these protocols and services are described in either GOSIP (or POSIT) or in a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, is used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required interim network services. The POSIT will serve as the base for developing the 50000 series, with DOD enhancements. Unique military standards will be used only when necessary.

Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829. DTMP Working Groups shall be responsible for DSP development and informal service and agency coordination; the DTMP Plenary shall be responsible for final review and approval.

This document is part of a set of interim DOD data communications protocol profiles based on the Internet protocol suite and is intended to support the interoperability of DOD communication networks, including connectivity with the Defense Data Network (DDN).

This DSP 2045-14502, Part 6 contains one normative annex and one informative annex:

Annex A (normative)	DSPICS Requirements List (DPRL) and Combat Net Radio (CNR) DOD Standardized Protocol Implementation Conformance Statement (DSPICS) PROFORMA
Annex B (informative)	CONCLUDING MATERIAL

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The Preparing Activity for this standard is the Data Communications Protocol Standards Technical Management Panel (DTMP). The custodians for this document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communications Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization
ATTN: DTMP Chairman, JEBBD
Squier Hall, Bldg 283
Fort Monmouth, New Jersey 07703-5613

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Introduction

This DSP is defined within the context of functional standardization, according to the principles specified by ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829. The context of functional standardization is one part of the overall field of Information Technology (IT) standardization activities covering base standards, profiles, and registration mechanisms. A profile defines a combination of base standards that perform a specific well-defined Information Technology function. Profiles standardize the selection of options and other variations in the base standards to promote system interoperability.

One of the most important roles for a DSP is to serve as the basis for the development of recognized tests. DSPs also guide implementations in developing systems that fit the needs of the US Department of Defense (DoD). DSPs are produced not simply to 'legitimize' a particular choice of base standards and options, but to promote real system interoperability . The development and widespread acceptance of tests based on this and other DSPs is crucial to the successful realization of this goal.

The base standards of this DSP may include Request For Comments (RFCs) designated as Official Internet Architecture Board (IAB) standards, other RFCs, and Open Systems Interconnection (OSI) Layer Standards from the OSI Reference Model.

This document is intended to be part of a complete TCP/IP based transport profile. The full profile specifies the use of the Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Internet Protocol (IP) services operating over the subnetwork, i.e. Layers 1, 2, and 3A. Part 6 applies to Combat Net Radio (CNR) as specified in MIL-STD-188-220A, Interoperability Standard For Digital Message Transfer Devices (DMTDs), and between DMTDs and applicable Command, Control, Communications, Computers, and Intelligence (C4I) systems at the subnetwork level. This DSP, Part 6, specifies use of a relay feature and a subnetwork independent convergence function (SNDCF) that operates below IP, and the complete services expected of Layers 1, 2, and 3A.

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Information Technology -DOD Standardized Profile - Internet Transport Profile for DOD Communications - Part 6: Combat Network Radio (CNR)

1 Scope

1.1 General

This DOD Standardized Profile (DSP) MIL-STD-2045-14502, Part 6 applies to the requirements for interoperability of Digital Message Transfer Devices (DMTDs), and between DMTDs and applicable Command, Control, Communications, Computers, and Intelligence (C4I) systems at the subnetwork level.

1.2 Position within the taxonomy

This profile is classified as MIL-STD-2045-14502, Part 6 in accordance with MIL-HDBK-829.

1.3 Scenario

This DSP specifies the provisions of the subnetwork layers of MIL-STD-188-220A, Interoperability Standard For Digital Message Transfer Device (DMTD) Subsystems. The layer standards which make up this profile are shown in Figure 1 in the shaded area.

ISO/TCP/IP Layers	Base Standard	Taxonomy
TRANSPORT LAYER	IAB STD 7 (TCP) IAB STD 6 (UDP) SELECTIVE DIRECTED BROADCAST - SEGMENTATION / REASSEMBLY	DSP TCP/IP Group Profile MIL-STD-2045-14502, Part 1A
NETWORK LAYER	IP/ICMP IAB STD 5 (IP)	
INTRANET LAYER	MIL-STD-188-220A, Paragraph 5.3 & 5.4	
DATA LINK LAYER	MIL-STD-188-220A, Paragraph 5.3 & 5.4	Combat Net Radio (CNR) Access
PHYSICAL LAYER	MIL-STD-188-220A, Paragraph 5.1 & 5.2	

Figure 1 - Profile Protocol Stack

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2. References

2.1 Government Documents

2.1.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

MIL-HDBK 829A	1 September 1994	*Guidelines for DOD Standardized Profiles*
MIL-STD-188-100 (Series)	17 November 1976	*Common Long Haul and Tactical Communications System Technical Standards*
MIL-STD-188-110A	30 September 1991	*Interoperability and Performance Standards for Data Modems*
MIL-STD-188-114	13 December 1991	*Electrical Characteristics of Digital Interface Circuits*
MIL-STD-188-220A	27 July 1995	*Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems*
FIPS 163 92	2 April 1992	*2400 Bits Per second Two Wire Duplex Modems for Data Communications Use on Telephone Type Circuits*
FIPS 164 92	2 April 1992	*2400 Bits Per second Four Wire Duplex Modems for Data Communications Use on Telephone Type Circuits*
FIPS-STD-1037	3 June 1991	*Glossary of Telecommunication Terms*

DOD activities may obtain copies of DOD directives through their own publication channels or from the DOD Single Stock Point, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other federal agencies and the public may purchase copies from the U.S. Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.

Copies of Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Others must request copies of FIPS from the National Technical Information Services, 5285 Port Royal, Springfield, VA 22161-2171.

2.1.2 Other Government documents, drawings and publications.

NONE

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2.2 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation.

2.2.1 Profiles

MIL-STD-2045-14502-1A	27 July 1995	*Transport and Internet Services*

2.2.2 Base Standards

MIL-STD-188-220A	27 July 1995	*Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems*

2.2.3 Other Non Government documents, drawings, and publications

International Standard, Information Processing Systems - Open Systems Interconnection - Internal Organization of the Network Layer, ISO 8648, dated 15 February 1988.

2.3 Order of precedence

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 Definitions

Terms used in this DSP are defined and discussed extensively in the referenced base standard. The following are provided for non-technical assistance purposes in this DSP, and may not be the same as that used in the base standard. Where different, the base standard takes precedence.

HAVEQUICK	UHF electronic protection waveform
INTRANET LAYER	A subset of the network layer that performs routing within a single subnetwork and extends link addressing by routing across one or more hops within a subnetwork. It corresponds to a Subnetwork Access Control Protocol (SnACP) as described in ISO 8648.

4 Symbols and abbreviations

Symbols used in this DSP are defined in the referenced base standard. The following are repeated for ease of use in this DSP.

DMTD	Digital Message Transfer Device
DSP	DOD Standardized Profile
ICOM	Integrated COMSEC
IP	Internet Protocol
SINCGARS	Single-Channel Ground-Air Radio System
SIP	SINCGARS Improvement Program
SNDCF	Subnetwork Dependent Convergence Function
TCP	Transmission Control Protocol

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5 Requirements

5.1 General Requirements

This DSP is based on the requirements for interoperability of Digital Message Transfer Devices (DMTDs), and between DMTDs and applicable Command, Control, Communications, Computers, and Intelligence (C4I) systems at the subnetwork level.

A conforming implementation of MIL-STD-2045-14502, Part 6 shall satisfy:

- all requirements in the remainder of this clause 5
- all mandatory requirements of the base standards referenced by this DSP
- all the constraints specified in Annex A (normative), DPRL
- all requirements of Annex A (normative), DSPICS Proforma

5.2 Intranet Layer

This DSP calls out an Intranet Layer feature that is used to accomplish relay between stations, and act as a subnetwork dependent convergence function (SNDCF). The relay feature is used to accomplish relay between stations within the same subnetwork that are not capable of receiving direct broadcast due to terrain or proximity. In support of this attribute, the Intranet Layer protocol also accomplishes a network topology and link connectivity update that provides constant refreshing of the location and connectivity of the net subscribers. The Intranet Layer protocol includes a military-defined feature that accomplishes selective relay to one or more destinations within the subnetwork when invoked. The relay feature is described in MIL-STD 188-220A, paragraph 5.4.1. MIL-STD-188-220A, Appendix I describes a procedure for accomplishing source directed relay using the relay feature. Implementations shall meet the requirements of Source Directed Relay as set forth in MIL-STD 188-220A, paragraph 5.4.1, and Appendix I.

The SNDCF operates over the relay feature described above and provides the functions needed to map address and type of service information between the IP and Data Link Layer protocol. The SNDCF is described in MIL-STD-188-220A, paragraph 5.4.2.

Conforming implementations shall meet the requirements of Annex A of this DSP.

5.2.1 Static Conformance Requirements

There are no additional static conformance requirements.

5.2.2 Dynamic Conformance Requirements

There are no additional dynamic conformance requirements.

5.3 Data Link Layer Requirements

Conforming implementations must satisfy the data link requirements specified in paragraphs 5.3 and 5.4 of Mil-Std 188-220A and Annex A of this DSP.

5.3.1 Static Conformance Requirements

There are no additional static conformance requirements.

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5.3.2 Dynamic Conformance Requirements

There are no additional dynamic conformance requirements.

5.4 Physical Layer Requirements

Conforming implementations must satisfy the physical layer requirements specified in paragraphs 5.1 and 5.2 of Mil-Std 188-220A and Annex A of this DSP.

5.4.1 Static Conformance Requirements

There are no additional static conformance requirements.

5.4.2 Dynamic Conformance Requirements

There are no additional dynamic conformance requirements.

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ANNEX A (normative)

DSPICS REQUIREMENTS LIST (DPRL) AND COMBAT NET RADIO DSPICS Proforma

A.1 Introduction

This document provides the DOD Standardized Profile Implementation Conformance Statements (DSPICS) Requirements List (DPRL) for implementations of MIL-STD-2045-14502, Part 6. The DSPICS for an implementation is generated by completing the DPRL in accordance with the following instructions.

An implementation shall satisfy the mandatory conformance requirements of the base standards referenced in this profile.

An implementation's completed DPRL is called the DSPICS. The DSPICS states which capabilities and options of the protocol have been implemented. The following can use the DSPICS:

(a) the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight.

(b) the supplier and acquirer or potential acquirer of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard DSPICS proforma.

(c) the user or potential user of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to internetwork can often be predicted from incompatible DSPICSs).

(d) a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.1.1 Notation

The following notations and symbols from MIL-HDBK 829A, which references ISO/IEC TR 10000-1 and -2, are used in the DPRL to indicate the status of features:

Status Symbols

m	- mandatory
m.<n>	- support of every item of the group labeled by the same numeral <n> required, but only one is active at a time
o	- optional
o.<n>	- optional, but support of at least one of the group of options labeled by the same numeral <n> is required
c	-conditional
-	- non-applicable (i.e. logically impossible in the scope of the profile)
x	-excluded or prohibited
i	-out of scope of profile (left as an implementation choice)

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In addition, the symbol "•" is used to indicate an option whose status is not constrained by the profile (status in the base standard). The o.<n> notation is used to show a set of selectable options (i.e., one or more of the set must be implemented) with the same identifier <n>

Two character combinations may be used for dynamic conformance requirements. In this case, the first character refers to the static (implementation) status, and the second refers to the dynamic (use); thus "mo" means "mandatory to be implemented, optional to be used."

Notations for Conditional Status

The following predicate notations are used:

<predicate>:: This notation introduces a group of items, all of which are conditional on <predicate>.

<predicate>: This notation introduces a single item which is conditional on <predicate>.

In each case, the predicate may identify a profile feature, or a boolean combination of predicates. ("^" is the symbol for logical negation.)

<index>. This predicate symbol means that the status following it applies only when the DSPICS states that the features identified by the index are supported. In the simplest case, <index> is the identifying tag of a single DSPICS item. The symbol <index> also may be a Boolean expression composed of several indices.

<index>:: When this group predicate is true, the associated clause should be completed.

Notations used in the Protocol Feature Column

<r> Symbol used to denote the receiving system.
<t> Symbol used to denote the transmitting system.

Support Column Symbols

The support of every item as claimed by the implementor is stated by circling the appropriate answer (Yes, No, or NA) in the support column:

Yes Supported by the implementation.
No Not supported by the implementation.
NA Not applicable.

Base standard requirements are shown using the equivalent notations in upper case (e.g., M, O, X).

A.1.2 Footnotes

Footnotes to the proforma are indicated by superscript numerals. The footnote appears on the page of the first occurrence of the numeral. Subsequent occurrences of a numeral refer to the footnote of the first occurrence.

A.1.3 Instructions for Completing the DPRL

A DSP implementor shows the extent of compliance to a DSP by completing the DPRL; that is, compliance to all mandatory requirements and the options that are not supported are shown. The resulting completed DPRL is called a DSPICS. Where this profile refines the features of the base standards, the requirements expressed in this DPRL shall be applied (as indicated in DPRL items with no "Profile Support" column) to constrain the allowable responses in the base standard DSPICS proforma. When this profile makes additional requirements, the "Profile Support" column for such DPRLs shall be completed. In this column, each response shall be selected either from the indicated set of

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responses, or it shall comprise one or more parameter values as requested. If a conditional requirement is inapplicable, use the Not Applicable (NA) choice. If a mandatory requirement is not satisfied, exception information must be supplied by entering a reference Xi, where i is a unique identifier, to an accompanying rationale for the noncompliance. When the profile requirement is expressed as a two-character combination (as defined in A.1.1 above), the response shall address each element of the requirement; e.g., for the requirement "mo," the possible compliant responses are "yy" or "yn."

A.1.4 Instructions for Completing the DSPICS (Support Column)

The supplier of a protocol implementation which is claimed to conform to the Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems, MIL-STD-188-220A, shall complete the following DOD Standardized Profile (DSP) Protocol Implementation Conformance Statement (DSPICS) proforma.

A completed DSPICS proforma is the DSPICS for the implementation in question. The DSPICS is a statement of which capabilities and options of the protocol have been implemented. The DSPICS can be used:

(a) By the protocol implementor, as a check-list to reduce the risk of failure to conform to the standard through oversight;

(b) By the supplier and acquirer - or potential acquirer - of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard DSPICS proforma;

(c) By the user - or potential user - of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to internetwork can often be predicted from incompatible DSPICSs);

(d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.1.4.1 General Structure of the DSPICS Proforma

The DSPICS Proforma - Implementation Identification and Protocol Summary - is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation. The DSPICS Proforma is usually a separate annex. However, in this DSP, the DSPICS has been combined with the basic DPRL since the implementation decisions required by the DSPICS from the DPRL can all be contained in the support column.

The main part of the DSPICS Proforma (and the DPRL in this case) is a fixed-format questionnaire divided into a number of major subclauses; these can be divided into further subclauses each containing a group of individual items. Answers to the questionnaire items are to be provided in the far right column by simply marking an answer (i.e. by circling the applicable entry) to indicate a restricted choice (Yes, No or NA) or by entering a value or a set of range of values.

The Structure of the DSPICS questionnaire consists of 4 main sections:

(1)	Subnetwork Dependent Convergence Function (SNDCF)	(Network Layer)
(2)	Intranet Protocol	(Network Layer)
(3)	Broadcast Protocol	(Data Link Layer)
(4)	Broadcast Protocol	(Physical Layer)

Each item is identified by an item identification number in the first column. The second column contains the statement of function or the question to be answered. The third column contains the reference to the material that specifies the item in the main body of the standard. The last two columns record the status of the item - whether support is mandatory, optional, prohibited or conditional - and provide the space for answers.

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A supplier may also provide - or be required to provide - further information, categorized as either Additional Information or Exceptional Information. When present, each kind of further information is to be provided in a further subclause of items labeled A<i> or X<i>, respectively, for cross-referencing purposes. The index <i> is any unambiguous identification for the item (e.g. a number); there are no other restrictions on format and presentation of additional information.

A completed DSPICS proforma, including any Additional Information or Exceptional Information, is the Protocol Implementation Conformance Statement for the implementation in question.

> Note: Where an implementation is capable of being configured in more than one way, a single DSPICS may be able to describe all such configurations; however, the supplier has the choice of providing more than one DSPICS, each covering some subset of the implementation's configuration capabilities, in cases where this makes for easier and clearer presentation of the information.

A.1.4.2 Additional Information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the DSPICS. It is not intended or expected that a large quantity will be supplied, and a DSPICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations; or a brief rationale - based perhaps upon specific application needs - for the exclusion of features which, although optional, are nonetheless commonly present in implementations of the protocol.

References to items of Additional Information may be entered next to any answer in the questionnaire, and may be included in items of Exception Information.

A.1.4.3 Exception Information

It may occasionally happen that a supplier will wish to answer an item with mandatory or prohibited status (after any conflicts with the indicated requirement). No pre-printed answer will be found in the Support column for this: instead, the supplier shall write the missing answer into the Support column, together with an Xi reference to an item of Exception Information, and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to MIL-STD-188-220A.

Note - The situation described above could occur if a defect in the Standard is reported, a correction for which could be expected to change the requirement NOT met by the implementation.

A.1.4.4 Conditional Items

The DSPICS proforma contains a number of conditional items. These are items for which the status - mandatory, optional or prohibited - that applies is dependent upon whether or not certain other items are supported, or upon the values supported for other items.

In many cases, whether or not the item applies at all is conditional, as well as the status when the item does apply.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" answer is selected. Otherwise, individual conditional items are indicated by one or more conditional symbols (on separate lines) in the Status column.

A conditional symbol is of the form "<predicate>:<x>" where "<predicate>" (abbreviated 'pred') is a predicate as described in A.1.4.5 below, and "<x>" is one of the status symbols M, O, O.<n> or X.

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If the value of the predicate in any line of a conditional item is true, the conditional item is applicable, and its status is that indicated by the status symbol following the predicate; the answer column is to be marked in the usual way. If the value of a predicate is false, the Not Applicable (NA) answer is to be marked in the relevant line. (Each line in a multi-line conditional item should be marked; at most one line will require an answer other than NA.)

A.1.4.5 Predicates

A predicate is one of the following:

(a) An item identification number for an item in the DSPICS proforma: the value of the predicate is true if the item is marked as supported, and is false otherwise; or

(b) A predicate name, for a predicate defined elsewhere in the DSPICS proforma; see below.

The definition for a predicate name is a boolean expression constructed by combining simple predicates, as at (a) or (b) above, using the boolean operators AND, OR and NOT, and parentheses, in the usual way. The value of such a predicate is true if the boolean expression evaluates to true when the item identification numbers are interpreted as at (a) above.

A.2 Implementation Identification

Supplier	
Contact point for queries about the DSPICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification (e.g. name(s) and version(s) of machines and/or operating systems, system name(s)	

NOTES

1. Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.

2. The terms Name(s) and Version(s) should be interpreted appropriately to correspond with a supplier's terminology (e.g. Type, Series, Model).

A.2.1 Protocol Summary

Identification of Protocol Specification	Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems, MIL-STD-188-220A.
Identification of amendments and corrigenda to the DSPICS Proforma	Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems, MIL-STD-188-220A. Am.: Corr.: Am.: Corr.: Am.: Corr.: Am.: Corr.: Am.: Corr.:
Protocol Version(s) supported	

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Identification of Protocol Specification	Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems, MIL-STD-188-220A.
Identification of amendments and corrigenda to the DSPICS Proforma	Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems, MIL-STD-188-220A. Am.: Corr.: Am.: Corr.: Am.: Corr.: Am.: Corr.: Am.: Corr.:
Have any Exception items been required? No[ ] Yes[ ] (The answer Yes means that the implementation does not conform to MIL-STD-188-220A.	

Date of Statement	

A.2.2 Standard Referenced

This profile specifies the provision of services in an End System and Intermediate System from which a standardized IP network service operating over combat net radio (CNR) broadcast circuits is available (or can be made available). This profile is militarized in that it makes changes to the underlying civil standards in order to accomplish the necessary military requirements for combat net radio. It uses the following military standard (and the underlying civil standards contained therein):

MIL-STD-188-220A	27 July 1995	*Military Standard, Interoperability Standard For Digital Message Transfer Device Subsystems*

A.3 CNR Requirements List

The following tables give the DPRL for MIL-STD-188-220A requirements and support required for the Combat Net Radio (CNR) Intranet Layer, Datalink Layer and Physical Layer. The table reference column refers to MIL-STD 188-220A unless noted otherwise. The support column describes the implementation.

A.3.1 Basic Requirements

Item (series)	Protocol Feature	Reference	Status	Support
302	Subnetwork Dependent Convergence Function	5.4.2	M	yes
300	Intranet Layer Protocol	5.4.1	M	yes
200	Data Link Layer	5.3	M	yes
100	Physical Layer	5.1 and 5..2	M	yes

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A.3.1.1 Subnetwork Dependent Convergence Function (SNDCF)

Item	Protocol Feature	Reference	Status	Support
302.1	Determine destination function	5.4.2.1	M	yes
302.2	Address mapping function	5.4.2.2	M	yes
302.3	Type of service function	5.4.2.3	M	yes
302.4	Intranet send request	5.4.2.4	M	yes

A.3.2 Intranet relay

Item	Protocol Feature	Reference	Status	Support
301.1	Intranet Header	5.4.1.1	M	yes
301.2	Topology Update	5.4.1.2 and Appendix B	MO	yes__NA__
301.3	Intranet Layer Interactions	5.4.1.3	M	yes
301.4	Source Directed Relay Procedure	Appendix I	MO	yes__NA__

A.3.2.1 Intranet header

Item	Protocol Feature	Reference	Status	Support
301.1.1	Version	5.4.1.1.1	M	yes
301.1.2	Message Type	5.4.1.1.2	M	yes
301.1.3	Intranet Header Length ≥ 3 octets	5.4.1.1.3	M	yes
301.1.4	Type of Service	5.4.1.1.4	M	yes
301.1.5	Message Identification Number	5.4.1.1.5	M	yes
301.1.6	Maximum Hop Count	5.4.1.1.6	M	yes
301.1.7	Destination/Relay Status Byte	5.4.1.1.7	M	yes
301.1.7a	Distance Subfield	5.4.1.1.7.1	M	yes
301.1.7b	REL bit	5.4.1.1.7.2	M	yes
301.1.7c	Source Directed (S/D) bits	5.4.1.1.7.3	M	yes
301.1.7d	DES bit	5.4.1.1.7.4	M	yes
301.1.7e	ACK bit	5.4.1.1.7.5	M	yes
301.1.8	Originator Address	5.4.1.1.8	M	yes
301.1.9	Destination/Relay Address	5.4.1.1.9	M	yes

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A.3.2.2 Topology update

Item	Protocol Feature	Reference	Status	Support
301.2.1	Topology update length less than MTU minus 8 octets	5.4.1.2.1	301.2:M	yes
301.2.2	Topology update ID (0-255)	5.4.1.2.2	301.2:M	yes
301.2.3	Node address	5.4.1.2.3	301.2:M	yes
301.2.4	Node status byte	5.4.1.2.4	301.2:M	yes
301.2.4.1	Link Quality	5.4.1.2.4.1	301.2:M	yes
301.2.4.2	Hop Length	5.4.1.2.4.2	301.2:M	yes
301.2.4.3	NR bit	5.4.1.2.4.3	301.2:M	yes
301.2.4.4	Quiet bit	5.4.1.2.4.4	301.2:M	yes
301.2.5	Node predecessor address	5.4.1.2.5	301.2:M	yes
301.2.6	Spanning Trees	B3.1	301.2:M	yes
301.2.7	Exchanging Spanning Trees	B4.1	301.2:M	yes
301.2.8	Sparse Spanning Trees	B4.3	301.2:M	yes
301.2.9	Rules for exchanging topology updates	B4.4	301.2:M	yes
301.2.10	Non-relayers	B4.5	301.2:M	yes
301.2.11	Quiet Nodes	B4.6	301.2:M	yes

A.3.2.3 Intranet Layer Data Transfer Interactions

Item	Protocol Feature	Reference	Status	Support
301.3.1	Topology Update Request Message	5.4.1.3, B.4.7	MO	yes NA
301.3.2	Intranet Layer (IL) Unitdata request parameters	5.4.1.4a	M	yes
301.3.3	Destination(s)	5.4.1.4a	M	yes
301.3.4	Source	5.4.1.4a	M	yes
301.3.5	Quality of Service	5.4.1.4a	M	yes
301.3.5.1	Precedence	5.4.1.4a	M	yes
301.3.5.2	Throughput - (Normal/High)	5.4.1.4a	M	yes
301.3.5.3	Delay - (Normal/Low)	5.4.1.4a	M	yes
301.3.5.4	Reliability - (Normal/High)	5.4.1.4a	M	yes
301.3.5.5	Data / Data Length	5.4.1.4a	M	yes
301.3.6	IL-Unitdata Indication parameters	5.4.1.4b	M	yes

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Item	Protocol Feature	Reference	Status	Support
301.3.6.1	Destination(s)	5.4.1.4b	M	yes
301.3.6.2	Source	5.4.1.4b	M	yes
301.3.6.3	Data / Data Length	5.4.1.4b	M	yes
301.3.7	IL-Status Indication	5.4.1.4b	M	yes
301.3.7.1	Acknowledgement Failure	5.4.1.4b	M	yes
301.3.7.2	Connection Status / Failure	5.4.1.4b	M	yes

A.3.2.4 Source Directed Relay Procedure

Item	Protocol Feature	Reference	Status	Support
301.4.1	Forward Routing	Appendix I,14.1	MO	yes__NA__
301.4.2	End-to-End Acknowledgments	Appendix I,14.2	MO	yes__NA__

A.3.3 Data Link Layer DPRL

A.3.3.1 Data link layer

Item	Protocol Feature	Reference	Status	Support
201	Transmission Header	5.3.1	M	yes
202	Net access control (NAC)	5.3.2	M	yes
203	Types of procedures	5.3.3	M	yes
204	Data link frame	5.3.4	M	yes
205	Operational parameters	5.3.5	M	yes
206	Commands and responses	5.3.6	M	yes
207	Description of procedures by type	5.3.7	M	yes
208	Data link initialization	5.3.8	M	yes
209	Frame transfer	5.3.9	M	yes

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Item	Protocol Feature	Reference	Status	Support
210	Flow control	5.3.10	M	yes
211	Acknowledgment and response	5.3.11	M	yes
212	Invalid frame	5.3.12	M	yes
213	Retransmission	5.3.13	M	yes
214	Error Detection and Correction (not used in packet mode)	5.3.14	MO	yes__NA__
215	Data Scrambling (not used in packet mode)	5.3.15	MO	yes__NA__
216	Link-layer interactions	5.3.16	M	yes

A.3.3.2 Transmission header

Item	Protocol Feature	Reference	Status	Support
201.1	Flag Sequence	5.3.4.2.1	M	yes
201.2	FEC Selection	5.3.1.1	M	yes
201.3	TDC Selection	5.3.1.1	M	yes
201.4	Scrambling Selection	5.3.1.1	M	yes
201.5	Transmission Queue	5.3.1.2	M	yes
201.6	T bits	5.3.1.2.1	M	yes
201.7	Queue Precedence	5.3.1.2.2	202.4.5:M	yes__NA__
201.8	Queue Length	5.3.1.2.3	202.4.5:M	yes__NA__
201.9	Data Link Precedence	5.3.1.2.4	202.4.6:M	yes__NA__
201.10	First Subscriber Number	5.3.1.2.5	202.4.6:M	yes__NA__
201.11	Frame check sequence	5.3.4.2.5	M	yes

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A.3.3.3 Net Access Control (NAC)

Item	Protocol Feature	Reference	Status	Support
202.1	Net busy sensing	Appendix C C3.1	M	yes
202.2	Response hold delay (RHD) procedures	Appendix C C3.2	M	yes
202.3	Timeout period (TP) procedures	Appendix C C3.3	M	yes
202.4	Net access delay (NAD) procedures	Appendix C C3.4	M	yes
202.4.1	Random net access delay (R-NAD)	Appendix C C3.4.1	M: < 202.4 >	yes__NA__
202.4.2	Prioritized net access delay (P-NAD)	Appendix C C3.4.2	M: < 202.4 >	yes__NA__
202.4.3	Hybrid net access delay (H-NAD)	Appendix C C3.4.3	M: < 202.4 >	yes__NA__
202.4.5	Radio Embedded network access delay (RE-NAD)[1]	Appendix C C3.4.4	M: < 202.4 >	yes__NA__
202.4.6	Deterministic Adaptive Prioritized network access delay (DAP-NAD)	Appendix C C3.4.5	M: < 202.4 >	yes__NA__
202.5	Net access timers	Appendix C, C3.2 - C3.4	M	yes
202.5.1	Response hold delay (RHD) timer	Appendix C, C3.2	M	yes
202.5.2	Timeout period (TP) timer	Appendix C, C3.3	M	yes
202.5.3	Net access delay (NAD) timer (not used with RE-NAD)	Appendix C, C3.3	M	yes
202.6	Scheduler	5.3.2.1	202.4.5:M	yes__NA__
202.6.1	Random Schedule interval	C3.4.4.1.1 C3.4.4.1.2 C3.4.4.1.3 C43.4.4.1.4	202.6:M	yes__NA__
02.6.2	Immediate mode scheduling	C3.4.4.1.5	202.6:M	yes__NA__
202.7	Voice/data net sharing	Appendix C, C3.5	101.1: O 101.6: X 101.7: O	101.1:y_n_ 101.6: no 101.7:y_n_

[1]1: RE NAD is physical layer specific RE-NAD cannot be used in those cases where the physical layer requirements are not met.

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A.3.3.4 Types of Procedures

Item	Protocol Feature	Reference	Status	Support
203.1	Type 1 operation (acknowledged and unacknowledged)	5.3.3.1.1	M	yes
203.2	Type 2 operation	5.3.3	O	yes __ no__
203.4	Type 4 operation	5.3.3	O	yes __ no__
203.5	Command and response features	5.3.3	M	yes

A.3.3.5 Types of frames

Item	Protocol Feature	Reference	Status	Support
204.1	Unnumbered frame (U-PDU)	5.3.4.1.1	M	yes
204.2	Information frame (I-PDU) Type 1 operations Type 2 operations Type 4 operations	5.3.4.1.2	 X 203.2:M X	 NA yes__NA__ NA
204.3	Supervisory frame(S-PDU) Type 1 operations Type 2 operations Type 4 operations	5.3.4.1.3	 X 203.2:M 203.4:M	 NA yes__NA__ yes__NA__

A.3.3.5.1 Data link frame structure

Item	Protocol Feature	Reference	Status	Support
204.5	Data link PDU construction procedures	5.3.4.3	M	yes
204.4.1	Flag sequence	5.3.4.2.1	M	yes
204.4.2	Address fields	5.3.4.2.2	M	yes
204.4.3	Control field	5.3.4.2.3	M	yes
204.4.4	Information field	5.3.4.2.4	M	yes
204.4.5	Frame check sequence	5.3.4.2.5	M	yes
204.5	Data link PDU construction procedures	5.3.4.3	M	yes

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A.3.3.5.2 Address fields

Item	Protocol Feature	Reference	Status		Support	
204.2.1.1	Address format	5.3.4.2.2.1	M		yes	
204.2.1.2	Addressing convention	5.3.4.2.2.2	M		yes	
204.2.1.1.1	Source address	5.3.4.2.2.2.1.1	M		yes	
204.2.1.1.2	Destination address(es)	5.3.4.2.2.2.1.2	M		yes	
204.2.1.3	Reserved address	5.3.4.2.2.2.2.1	M		yes	
204.2.1.4	Special addresses	5.3.4.2.2.2.2.2	Send: M	Recv: M	Send: yes_ no_	Recv: yes
204.2.1.5	Individual addresses	5.3.4.2.2.2.2.3	M		yes	
204.2.1.6	Group multicast addresses	5.3.4.2.2.2.2.4	Send: O	Recv: M	Send: yes_ no_	Recv: yes
204.2.1.7	Individual and multicast addresses mixed	5.3.4.2.2.2.2.5	M		yes	
204.2.1.8	Global multicast addressing	5.3.4.2.2.2.2.6	M		yes	
204.2.1.9	Mapping	5.3.4.2.2.3	M		yes	

A.3.3.5.3 Control field

Item	Protocol Feature	Reference	Status	Support
204.4.3.1	Type 1 operations(8 bit pattern)	5.3.4.2.3.1	M	yes
204.4.3.2	Type 2 operations(8 or 16 bit pattern)	5.3.4.2.3.2	203.2:M	yes__NA__
204.4.3.3	Type 4 operations (16 bit pattern)	5.3.4.2.3.3	203.4:M	yes__NA__
204.4.3.4	Poll/final bit Type 1 Type 2 Type 4	5.3.4.2.3.4	 M 203.2:M X	 yes yes__NA__ NA
204.4.3.5	Sequence numbers (Type 2 only)	5.3.4.2.3.5	203.2:M	yes__NA__
204.4.3.6	Identification numbers (Type 4 only)	5.3.4.2.3.6	203.4:M	yes__NA__
204.4.3.7	Precedence (Type 4 only)	5.3.4.2.3.7	203.4:M	yes__NA__

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A.3.3.5.4 Data link PDU construction

Item	Protocol Feature	Reference	Status	Support
204.5.1	Order of bit transmission	5.3.4.3.1	M	yes
204.5.2	Zero-bit insertion algorithm	5.3.4.3.2	M	yes

A.3.4 Operational parameters

Item	Protocol Feature	Reference	Status	Support
205.1	Type 1 operational parameters	5.3.5.1	M	yes
205.2	Type 2 operational parameters	5.3.5.2	203.2:M	yes__ NA__
205.3	Type 4 operational parameters	5.3.5.3	203.4:M	yes__ NA__
205.3.1	Identification Number	5.3.5.3.1	203.4:M	yes__ NA__

A.3.4.1 Type 2 operational parameters

Item	Protocol Feature	Reference	Status	Support
205.2.1	Modulus	5.3.5.2.1	203.2:M	yes__ NA__
205.2.2	PDU state variables and sequence numbers	5.3.5.2.2	203.2:M	yes__ NA__
205.2.3	Send state variable V(S)	5.3.5.2.2.1	203.2:M	yes__ NA__
205.2.4	Send sequence number N(S)	5.3.5.2.2.2	203.2:M	yes__ NA__
205.2.5	Receive state variable V(R)	5.3.5.2.2.3	203.2:M	yes__ NA__
205.2.6	Receive sequence number N(R)	5.3.2.2.2.4	203.2:M	yes__ NA__
205.2.7	Poll/final (P/F) bit	5.3.5.2.3	203.2:M	yes__ NA__
205.2.8	Poll bit functions	5.3.5.2.3.1	203.2:M	yes__ NA__
205.2.9	Final bit functions	5.3.5.2.3.2	203.2:M	yes__ NA__

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A.3.4.2 Commands and Responses

Item	Protocol Feature	Reference	Status	Support
206.1	Type 1 commands and responses	5.3.6.1	M	yes
206.2	Type 2 commands and responses	5.3.6.2	203.2:M	yes__NA__
206.3	Type 4 commands and responses	5.3.6.3	203.4:M	yes__NA__

A.3.4.2.1 Type 1 commands and responses

Item	Protocol Feature	Reference	Status	Support	
206.1.1	Unnumbered information (UI) command	5.3.6.1.1	M	yes	
206.1.2	Unnumbered receive ready (URR) command	5.3.6.1.2	M	yes	
206.1.3	Unnumbered receive not ready (URNR) command	5.3.6.1.3	M	yes	
206.1.4	Exchange identification (XID) command	5.3.6.1.4	MO	Send: yes__ no__	Recv: yes
206.1.5	Test command	5.3.6.1.5	M	yes	
206.1.6	Unnumbered receive ready (URR) response	5.3.6.1.6	M	yes	
206.1.7	Exchange identification (XID) response	5.3.6.1.7	M	yes	
206.1.8	Test response	5.3.6.1.8	M	yes	
206.1.9	Unnumbered receive not ready (URNR) response	5.3.6.1.9	M	yes	

A.3.4.2.2 Type 2 commands and responses

Item	Protocol Feature	Reference	Status	Support
206.2.1	Information transfer format command and response	5.3.6.2.1	203.2:M	yes__NA__
206.2.2	Supervisory format commands and responses	5.3.6.2.2	203.2:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
206.2.3	Receive ready (RR) command and response	5.3.6.2.2.1	203.2:M	yes__NA
206.2.4	Reject (REJ) command and response	5.3.6.2.2.2	203.2:M	yes__NA__
206.2.5	Receive not ready (RNR) command and response	5.3.6.2.2.3	203.2:M	yes__NA__
206.2.6	Selective reject (SREJ) command and response	5.3.6.2.2.4	203.2:M	yes__NA__
206.2.7	Unnumbered format commands and responses	5.3.6.2.3	203.2:M	yes__NA__
206.2.7.1	Set asynchronous balanced mode (SABME) command	5.3.6.2.3.1	203.2:M	yes__NA__
206.2.7.2	Disconnect (DISC) command	5.3.6.2.3.2	203.2:M	yes__NA__
206.2.7.3	Reset (RSET) command	5.3.6.2.3.3	203.2:M	yes__NA__
206.2.7.4	Unnumbered acknowledgment (UA) response	5.3.6.2.3.4	203.2:M	yes__NA__
206.2.7.5	Disconnect mode (DM) response	5.3.6.2.3.5	203.2:M	yes__NA__
206.2.7.6	Frame reject (FRMR) response	5.3.6.2.3.6	203.2:M	yes__NA__

A.3.4.2.3 Type 4 commands and responses

Item	Protocol Feature	Reference	Status	Support
206.3.1	Unnumbered information (DIA) transfer format command	5.3.6.3.1	203.4:M	yes__NA__
206.3.1.1	DIA PDU acknowledgement	5.3.6.3.1.1	203.4:M	yes__NA__
206.3.2	Supervisory format commands and responses	5.3.6.3.2	203.4:M	yes__NA__

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A.3.5 Description of procedures by type

Item	Protocol Feature	Reference	Status	Support
207.1	Description of Type 1 procedures	5.3.7.1	M	yes
207.2	Description of Type 2 procedures	5.3.7.2	203.2:M	yes__NA__
207.3	Description of Type 4 procedures	5.3.7.3	203.4:M	yes__NA__

A.3.5.1 Description of Type 1 procedures

Item	Protocol Feature	Reference	Status	Support
207.1.1	Modes of operation	5.3.7.1.1	M	yes
207.1.2	Procedure for addressing	5.3.7.1.2	M	yes
207.1.3	Procedure for the use of the P/F bit	5.3.7.1.3	M	yes
207.1.4	Procedures for logical data link set-up and disconnection	5.3.7.1.4	M	yes
207.1.5	Procedures for information transfer	5.3.7.1.5	M	yes

A.3.5.1.1 Type 1 procedures for information transfer

Item	Protocol Feature	Reference	Status	Support
207.1.5.1	Sending UI command PDUs	5.3.7.1.5.1	M	yes
207.1.5.2	Receiving UI command PDUs	5.3.7.1.5.2	M	yes
207.1.5.3	Sending URR response PDUs	5.3.7.1.5.3	M	yes
207.1.5.4	Sending URNR response PDUs	5.3.7.1.5.4	M	yes
207.1.5.5	Receiving UI acknowledgment	5.3.7.1.5.5	M	yes
207.1.5.6	Sending URNR command PDUs	5.3.7.1.5.6	M	yes
207.1.5.7	Receiving URNR command PDU	5.3.7.1.5.7	M	yes
207.1.5.8	Sending URR command PDUs	5.3.7.1.5.8	M	yes
207.1.5.9	Receiving URR command PDU	5.3.7.1.5.9	M	yes

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Item	Protocol Feature	Reference	Status	Support
207.1.5.10	Using XID command and response PDUs	5.3.7.1.5.10	M	yes
207.1.5.11	Using TEST command and response PDUs	5.3.7.1.5.11	M	yes

A.3.5.2 Description of Type 2 procedures

Item	Protocol Feature	Reference	Status	Support
207.2.1.1	Operational mode	5.3.7.2.1.1	203.2:M	yes__NA__
207.2.1.2	Non-operational mode	5.3.7.2.1.2	203.2:M	yes__NA__
207.2.2	Procedure for addressing	5.3.7.2.2	203.2:M	yes__NA__
207.2.3	Procedures for the use of the P/F bit	5.3.7.2.3	203.2:M	yes__NA__
207.2.4	Procedures for data link set-up and disconnection	5.3.7.2.4	203.2:M	yes__NA__
207.2.5	Procedures for information transfer	5.3.7.2.5	203.2:M	yes__NA__
207.2.6	Procedures for mode resetting	5.3.7.2.6	203.2:M	yes__NA__
207.2.6.1	Receiver action	5.3.7.2.6.1	203.2:M	yes__NA__
207.2.6.2	Initiator action	5.3.7.2.6.2	203.2:M	yes__NA__
207.2.6.3	Resetting with the FRMR PDU	5.3.7.2.6.3	203.2:M	yes__NA__
207.2.7	Procedures for sequence number resetting	5.3.7.2.7	203.2:M	yes__NA__
207.2.8	FRMR exception conditions	5.3.7.2.8	203.2:M	yes__NA__

A.3.5.2.1 Type 2 procedures for data link set-up and disconnection

Item	Protocol Feature	Reference	Status	Support
207.2.4.1	Data link connection phase	5.3.7.2.4.1	203.2:M	yes__NA__
207.2.4.1.1	Initiator Action	5.3.7.2.4.1.1	203.2:M	yes__NA__
207.2.4.1.2	Respondent Action	5.3.7.2.4.1.2	203.2:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
207.2.4.2	Information transfer phase	5.3.7.2.4.2	203.2:M	yes__NA__
207.2.4.3	Data link disconnection phase	5.3.7.2.4.3	203.2:M	yes__NA__
207.2.4.4	Data link disconnected phase	5.3.7.2.4.4	203.2:M	yes__NA__
207.2.4.5	Contention of unnumbered mode setting command PDUs	5.3.7.2.4.5	203.2:M	yes__NA__

A.3.5.2.2 Type 2 procedures for information transfer

Item	Protocol Feature	Reference	Status	Support
207.2.5.1	Sending I PDUs	5.3.7.2.5.1	203.2:M	yes__NA__
207.2.5.2	Receiving an I PDU	5.3.7.2.5.2	203.2:M	yes__NA__
207.2.5.3	Receiving of incorrect PDUs	5.3.7.2.5.3	203.2:M	yes__NA__
207.2.5.4	Receiving of out-of-sequence PDUs	5.3.7.2.5.4	203.2:M	yes__NA__
207.2.5.4.1	Reject response	5.3.7.2.5.4.1	203.2:M	yes__NA__
207.2.5.4.2	Selective reject response	5.3.7.2.5.4.2	203.2:M	yes__NA__
207.2.5.5	Receiving acknowledgment	5.3.7.2.5.5	203.2:M	yes__NA__
207.2.5.6	Receiving a SREJ PDU	5.3.7.2.5.6	203.2:M	yes__NA__
207.2.5.7	Receiving a RSET PDU	5.3.7.2.5.7	203.2:M	yes__NA__
207.2.5.8	Receiving a REJ PDU	5.3.7.2.5.8	203.2:M	yes__NA__
207.2.5.9	Receiving an RNR PDU	5.3.7.2.5.9	203.2:M	yes__NA__
207.2.5.10	Station busy condition	5.3.7.2.5.10	203.2:M	yes__NA__
207.2.5.11	Waiting acknowledgment	5.3.7.2.5.11	203.2:M	yes__NA__

A.3.5.3 Description of Type 4 procedures

Item	Protocol Feature	Reference	Status	Support
207.3.1	Modes of operation	5.3.7.3.1	203.4:M	yes__NA__
207.3.2	Procedure for addressing	5.3.7.3.2	203.4:M	yes__NA__
207.3.3	Procedure for the use of the P/F bit	5.3.7.3.3	203.4:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
207.3.4	Procedures for logical data link set-up and disconnection	5.3.7.3.4	203.4:M	yes__NA__
207.3.5	Procedures for information transfer	5.3.7.3.5	203.4:M	yes__NA__

A.3.5.3.1 Type 4 procedures for information transfer

Item	Protocol Feature	Reference	Status	Support
207.3.5.1	Sending DIA command PDUs	5.3.7.3.5.1	203.4:M	yes__NA__
207.3.5.2	Sending DRNR command PDU	5.3.7.3.5.2.1	203.4:M	yes _NA__
207.3.5.3	Receiving DRNR command PDU	5.3.7.3.5.2.2	203.4:M	yes__NA__
207.3.5.4	Sending DRNR response PDU	5.3.7.3.5.2.3	203.4:M	yes__NA__
207.3.5.4.1	Receiving DRNR response PDU	5.3.7.3.5.2.4	203.4:M	yes__NA__
207.3.5.5	Sending a DRR command PDU	5.3.7.3.5.3.1.1	203.4:M	yes_ NA
207.3.5.5.1	Sending a DRR response PDU	5.3.7.3.5.3.1.2	203.4:M	yes__NA__
207.3.5.5.2	Receiving DRR response PDU	5.3.7.3.5.3.2	203.4:M	yes__NA__

A.3.6 Data link initialization

Item	Protocol Feature	Reference	Status	Support
208.a	Octet range in UI, I, DIA PDUs: (708-3345)	5.3.8.1	M	yes
208.1	Type 1 logical data link parameters	5.3.8.1.1	M	yes
208.2	Type 2 data link parameters	5.3.8.1.2	203.2:M	yes__NA__
208.3	Type 4 data link parameters	5.3.8.1.3	203.4:M	yes__NA__
208.4	Exchange identification (XID) messages	5.3.8 and Appendix E	O	yes__no__
208.5	Network Configuration Operating Parameters	5.3.8 and Appendix E	O	yes__no__

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A.3.6.1 Type 1 logical data link parameters

Item	Protocol Feature	Reference	Status	Support
208.1.1	Acknowledgment timer	5.3.8.1.1.a, C3.3	M	yes
208.1.2	Busy-state timer	5.3.8.1.1.b	M	yes
208.1.3	Maximum number of transmissions, N4	5.3.8.1.1.c	M	yes
208.1.4	Minimum number of octets in a PDU	5.3.8.1.1.d	M	yes

A.3.6.2 Type 2 data link parameters

Item	Protocol Feature	Reference	Status	Support
208.2.1	Acknowledgment timer	5.3.8.1.2.a	203.2:M	yes__NA__
208.2.2	P-bit timer	5.3.8.1.2.b	203.2:M	yes__NA__
208.2.3	Reject timer	5.3.8.1.2.c	203.2:M	yes__NA__
208.2.4	Maximum number of transmissions, N2	5.3.8.1.2.d	203.2:M	yes__NA__
208.2.5	Maximum number of outstanding I PDUs,	5.3.8.1.2.e	203.2:M	yes__NA__
208.2.6	Minimum number of octets in a PDU	5.3.8.1.2.f	203.2:M	yes__NA__

A.3.6.3 Type 4 data link parameters

Item	Protocol Feature	Reference	Status	Support
208.3.1	Acknowledgement (T1) timer	5.3.8.1.3.a	203.4:M	yes__NA__
208.3.2	Maximum transmission attempts, N2	5.3.8.1.3.b	203.4:M	yes__NA__
208.3.3	k maximum number of outstanding DIA frames	5.3.8.1.3.c	203.4:M	yes__NA__

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A.3.6.4 Exchange identification XID

Item	Protocol Feature	Reference	Status	Support
208.4.1	XID format	Appendix E, E3.1	208.4:M	yes__NA__
208.4.2	XID command/response messages	Appendix E, E3.2	208.4:M	yes__NA__
208.4.3	XID poll/final bit	Appendix E, E3.3	208.4:M	yes__NA__
208.4.4	MIL-STD-188-220 parameter negotiation	Appendix E, E4.1	208.4:M	yes__NA__
208.4.5	Link Management messages	Appendix E, Tables E-2 thru E-7, para E4.1.1	208.4:M	yes__NA__
208.4.5.1	Join the network messages	Appendix E, E4.1.2	208.4:M	yes__NA__
208.4.5.2	Network Control Station	Appendix E, E4.1.2.1	O	yes__no__
208.4.5.3	Message exchange sequence	Appendix E, E4.1.2.2	O	yes__no__
208.4.5.4	Hello message	Appendix E, E4.1.3	O	yes__no__
208.4.5.5	Goodbye message	Appendix E, E4.1.4	O	yes__no__
208.4.6	Net-access-delay method	Appendix E, E5.1	208.4:M	yes__NA__

A.3.6.5 Network Configuration Operating Procedures

Item	Protocol Feature	Reference	Status	Support
208.5a	Group Address	Appendix E	208.5:M	yes__NA__
208.5.1	Link Address	Appendix E	208.5:M	yes__NA__
208.5.2	Station Class	Appendix E	208.5:M	yes__NA__
208.5.3	Robust Frame Format	Appendix E	208.5:M	yes__NA__
208.5.4	EDC Mode	Appendix E	208.5:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
208.5.5	Net busy detect time	Appendix E	208.5:M	yes__NA__
208.5.6	Equipment preamble time	Appendix E	208.5:M	yes__NA__
208.5.7	Keytime	Appendix E	208.5:M	yes__NA__
208.5.8	Equipment postamble time	Appendix E	208.5:M	yes__NA__
208.5.9	Carrier dropout time	Appendix E	208.5:M	yes__NA__
208.5.10	Equipment lag time	Appendix E	208.5:M	yes__NA__
208.5.11	Tolerance time	Appendix E	208.5:M	yes__NA__
208.5.12	Processing time	Appendix E	208.5:M	yes__NA__
208.5.13	NAD method	Appendix E	208.5:M	yes__NA__
208.5.14	Subscriber rank	Appendix E	208.5:M	yes__NA__
208.5.15	Number of stations	Appendix E	208.5:M	yes__NA__
208.5.16	Urgent Percent	Appendix E	208.5:M	yes__NA__
208.5.17	Priority percent	Appendix E	208.5:M	yes__NA__
208.5.18	Traffic load	Appendix E	208.5:M	yes__NA__
208.5.19	Max transmit seconds	Appendix E	208.5:M	yes__NA__
208.5.20	Physical concatenation	Appendix E	208.5:M	yes__NA__
208.5.21	Data link concatenation	Appendix E	208.5:M	yes__NA__
208.5.22	Max UI, DIA and I Info Bytes	Appendix E	208.5:M	yes__NA__
208.5.23	Type 2 ACK timer	Appendix E	208.5:M	yes__NA__
208.5.24	Type 2 k window	Appendix E	208.5:M	yes__NA__
208.5.25	Type 4 ACK timer	Appendix E	208.5:M	yes__NA__
208.5.26	Type 4 k window	Appendix E	208.5:M	yes__NA__
208.5.27	Response mode	Appendix E	208.5:M	yes__NA__

A.3.7 Frame transfer

Item	Protocol Feature	Reference	Status		Support	
209.1	PDU transmission	5.3.9.1	M		yes	
209.2	Data link concatenation	5.3.9.2	Send: O	Recv: M	Send: yes__ no__	Recv: yes

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Item	Protocol Feature	Reference	Status		Support	
209.2 a	Frame tx time < Network allowable	5.3.9.2	M	-	yes	-
209.3	Physical layer concatenation (all but RE-NAD)	5.3.9.3	Send: O	Recv: M	Send: yes__ no__	Recv: yes
209.4	PDU transmissions	5.3.9.4	M		yes	

A.3.8 Flow control

Item	Protocol Feature	Reference	Status	Support
210.1	Type 1 flow control	5.3.10.1	M	yes
210.2	Type 2 flow control	5.3.10.2	203.2:M else X	yes
210.3	Type 4 flow control	5.3.10.3	203.4:M else X	yes

A.3.9 Acknowledgment

Item	Protocol Feature	Reference	Status	Support
211.1	Type 1 acknowledgment	5.3.11.1.1	M	yes
211.2	Type 2 acknowledgment	5.3.11.1.2	203.2:M else X	yes
211.3	Type 4 acknowledgment	5.3.11.1.3	203.4:M else X	yes
211.4	Response mode	5.3.11.2	M	yes

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Class B devices and shall consist of the tests, conditions, and limits

A.3.10 Error detection and correction

Item	Protocol Feature	Reference	Status	Support
214.1	Forward-error-correction coding (not used in packet mode)	5.3.14.1 and Appendix F	MO	yes__NA__
214.2	Forward-error-correction preprocessing (not used in packet mode)	5.3.14.2	MO	yes__NA__
214.3	Time-dispersal coding (not used in packet mode)	5.3.14.3	MO	yes__NA__

A.3.11 Link Layer Interactions

Item	Protocol Feature	Reference	Status	Support
216.1	Datalink Layer (DL) Unitdata request parameters	5.3.16a	M	yes
216.1.1	Destination(s)	5.3.16a	M	yes
216.1.2	Source	5.3.16a	M	yes
216.1.3	Quality of Service	5.3.16a	M	yes
216.1.4	Precedence	5.3.16a	M	yes
216.1.5	Throughput - (Normal/High)	5.3.16a	M	yes
216.1.6	Delay - (Normal/High)	5.3.16a	M	yes
216.1.7	Reliability - (Normal/High)	5.3.16a	M	yes
216.1.8	Data / Data Length	5.3.16a	M	yes
216.2	DL-Unitdata Indication parameters	5.3.16b	M	yes
216.2.1	Destination(s)	5.3.16b	M	yes
216.2.2	Source	5.3.16b	M	yes
216.2.3	Data / Data Length	5.3.16b	M	yes
216.3	DL-Status Indication	5.3.16b	M	yes
216.3.1	Acknowledgement Failure	5.3.16b	M	yes
216.3.2	Connection Status / Failure	5.3.16b	M	yes

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A.4 Physical Layer DPRL

A.4.1 Interoperability standard for DMTD subsystems (Physical Layer)

Item	Protocol Feature	Reference	Status	Support
101	Physical layer transmission channel interfaces	5.1	M	yes
102	Physical layer protocol	5.2	M	yes

A.4.1.1 Physical layer transmission channel interfaces

Item	Protocol Feature	Reference	Status	Support
101.1	Non-return-to-zero (NRZ) interface	5.1.1.1	O.<3>	yes__NA__
101.2	Frequency-shift keying interface for voice frequency channels	5.1.1.2	O	yes no
101.3	Frequency-shift keying interface for single-channel radio	5.1.1.3	O	yes no
101.4	Conditioned diphase interface	5.1.1.4	O	yes no
101.5	Differential phase-shift keying interface for voice frequency channels	5.1.1.5	O	yes no
101.6	Packet Mode Interface	5.1.1.6	O.<3>	yes__NA__
101.7	Amplitude Shift Keying	5.1.1.7	O.<3>	yes__NA__

A.4.1.2 Non-Return-to-Zero (NRZ) interface

Item	Protocol Feature	Reference	Status	Support
101.1.1	NRZ unbalanced waveform conforms to paragraph 5.1.1.7 of MIL-STD-188-114A	5.1.1.1.1	101.1:M	yes__NA__
101.1.2	NRZ balanced waveform conforms to paragraph 5.2.1.7 of MIL-STD-188-114A	5.1.1.1.1	101.1:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
101.1.3	Output transmission rates shall be the following bit rates: 75, 150, 300, 600, 1200, 2400, 4800, 9600, and 16000 bits per second	5.1.1.1.2	101.1:M	yes__NA__
101.1.4	NRZ shall support half-duplex transmission	5.1.1.1.3	101.1:M	yes__NA__

A.4.1.3 Frequency-shift keying interface for voice frequency channels

Item	Protocol Feature	Reference	Status	Support
101.2.1	Frequency-shift keying (FSK) data modem characteristics shall conform to paragraph 5.2.2 of MIL-STD-188-110	5.1.1.2	101.2:M	yes__NA__
101.2.2	FSK modulation waveform shall conform to paragraph 5.2.2.1 of MIL-STD-188-110	5.1.1.2.1	101.2:M	yes__NA__
101.2.3	Characteristic frequencies of FSK interface for voice frequency channels of 600 bits per second or less shall be 1300 Hz for mark frequency and 1700 Hz for space frequency	5.1.1.2.1	101.2:M	yes__NA__
101.2.4	Characteristic frequencies of FSK interface for voice frequency channels of 1200 bits per second shall be 1300 Hz for mark frequency and 2100 Hz for space frequency	5.1.1.2.1	101.2:M	yes__NA__
101.2.5	Output transmission rates of the FSK interface shall be the following bit rates: 75, 150, 300, 600, and 1200 bits per second	5.1.1.2.2	101.2:M	yes__NA__
101.2.6	FSK interface shall support half-duplex transmission	5.1.1.2.3	101.2:M	yes__NA__

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A.4.1.4 Frequency-shift keying interface for single-channel radio

Item	Protocol Feature	Reference	Status	Support
101.3.1	FSK interface data modem characteristics shall conform to paragraph 5.1 of MIL-STD-188-110	5.1.1.3	101.3:M	yes__NA__
101.3.2	FSK modulation waveform shall conform to paragraph 5.1.1 and 5.1.2 of MIL-STD-188-110	5.1.1.3.1	101.3:M	yes__NA__
101.3.3	Characteristic frequencies of FSK interface for single channel radio shall be 1575 Hz for mark frequency and 2425 Hz for space frequency	5.1.1.3.1	101.3:M	yes__NA__
101.3.4	Output transmission rates of the single-channel FSK interface shall be the following bit rates: 600 and 1200 bits per second	5.1.1.3.2	101.3:M	yes__NA__
101.3.5	Single-channel FSK interface shall support half-duplex transmission	5.1.1.3.3	101.3:M	yes__NA__

A.4.1.5 Conditioned diphase interface

Item	Protocol Feature	Reference	Status	Support
101.4.1	A conditioned diphase (CDP) modulation data modem shall be used for this interface	5.1.1.4	101.4:M	yes__NA__
101.4.2	CDP modulation waveform shall conform to paragraph 5.4.1.4 of MIL-STD-188-200	5.1.1.4.1	101.4:M	yes__NA__
101.4.3	Unbalanced signal waveform conforms to paragraph 5.1.1.7 of MIL-STD-188-114	5.1.1.4.1	101.4:M	yes__NA__
101.4.4	Balanced signal waveform conforms to paragraph 5.2.1.7 of MIL-STD-188-114	5.1.1.4.1	101.4:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
101.4.5	Output transmission rate of the CDP interface shall be 16 kbps and 32kbps	5.1.1.4.2	101.4:M	yes__NA__
101.4.6	CDP interface shall support half-duplex transmission	5.1.1.4.3	101.4:M	yes__NA__

A.4.1.6 Differential phase-shift keying interface for voice frequency channels

Item	Protocol Feature	Reference	Status	Support
101.5.1	Differential phase-shift keying (DPSK) modulation data modem (2400 bps) and the phase-shift keying (PSK) modulation data modem (1200 bps) characteristics shall conform to the applicable requirements of MIL-STD-118-110A.	5.1.1.5	101.5: M	yes__NA__
101.5.2	DPSK modulation waveform shall conform to Appendix A of MIL STD-118-110A.	5.1.1.5.1	101.5: M	yes__NA__
101.5.3	PSK modulation waveform conforms to paragraph 5.3 of MIL-STD-118-110A.	5.1.1.5.1	101.5: M	yes__NA__
101.5.4	Output transmission rate of the DPSK interface shall be 2400 bps	5.1.1.5.2	101.5: M	yes__NA__
101.5.5	Output transmission rate of the PSK interface shall be 1200 bps	5.1.1.5.2	101.5: M	yes__NA__
101.5.6	DPSK and PSK interfaces shall support half-duplex transmission	5.1.1.5.3	101.5: M	yes__NA__

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A.4.1.7 Packet mode interface

Item	Protocol Feature	Reference	Status	Support
101.6.1	The packet mode interface shall use a modified CCITT X.21 half-duplex synchronous interface for transferring digital data across the interface between DTE/DCE	5.1.1.6	101.6:M	yes__NA__
101.6.2	The packet interface shall be a modified CCITT X.21 interface with a CCITT V.10 electrical circuit between the DTE/DCE (radio).	5.1.1.6	101.6:M	yes__NA__
101.6.3	The electrical characteristics of the packet mode interface shall be identical to CCITT V.10 for interfaces to voice-band modems.	5.1.1.6.1	101.6:M	yes__NA__
101.6.3a	The DTE shall accept DCE (radio) signal timing at 16 kbps.	5.1.1.6.2	101.6:M	yes__NA__
101.6.3b	The DTE shall be able to synchronize to the DCE timing and accept data at the signalling timing rate	5.1.1.6.2	101.6:M	yes__NA__
101.6.4	The packet mode interface shall support half-duplex transmission	5.1.1.6.3	101.6:M	yes__NA__

A.4.1.8 Amplitude Shift Keying

Item	Protocol Feature	Reference	Status	Support
101.7.1	The ASK waveform is a band-limited NRZ waveform with average white gaussian noise added to it. The ASK signal shall be a bipolar signal nominally centered around ground.	5.1.1.7.1	101.7:M	yes__NA__
101.7.2	The ASK signal-to-noise ratio shall be in the range of 0-12 decibels.	5.1.1.7.1	101.7:M	yes__NA__
101.7.3	The ASK signal shall be demodulated using an optimal bit synchronizer with a bit error rate performance of 1.5 decibels from theoretical	5.1.1.7.1	101.7:M	yes__NA__

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Item	Protocol Feature	Reference	Status	Support
101.7.4	The output transmission rates of the ASK interface shall be the following bit rates: 2400, 4800, 9600 and 16000 bits per second.	5.1.1.7.2	101.7:M	yes__NA__
101.7.5	The ASK interfaces shall support half-duplex transmission.	5.1.1.7.3	101.7:M	yes__NA__

A.4.2 Physical layer protocol

Item	Protocol Feature	Reference	Status	Support
102.1	Physical layer protocol data unit (PDU)	5.2.1	M	yes
102.2	Keytime delay (does not apply to packet mode)	5.2.2 and Appendix C, C30.2.b	M	yes
102.3	Net access control related indications (does not apply to packet mode)	5.2.3 and Appendix C, C30.1	M	yes
102.4	Physical layer to upper layer interactions	5.2.4	M	yes

A.4.2.1 Physical layer protocol data unit

Item	Protocol Feature	Reference	Status	Support
102.1.1	Physical-layer protocol data unit	5.2.1	M	yes
102.1.2	Communications security preamble and postamble	5.2.1.1 and Appendix D	M	yes
102.1.3	Asynchronous mode transmission synchronization field	5.2.1.2.1	101.7:M	yes__NA__
102.1.4	Synchronous Mode Transmission synchronization field	5.2.1.2.2	101.1:M	yes__NA__
102.1.5	Packet mode transmission synchronization field	5.2.1.2.3	101.6:M	yes__NA__
102.1.6	Data field	5.2.1.3	M	yes

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A.4.3 Asynchronous mode transmission synchronization field

Item	Protocol Feature	Reference	Status	Support
102.1.3.1	Bit synchronization subfield	5.2.1.2.1.1	102.1.3:M	yes__NA__
102.1.3.2	Robust frame synchronization subfield	5.2.1.2.1.2	102.1.3:M	yes__NA__
102.1.3.3	Robust frame format	N/A	102.1.3:M	yes__NA__
102.1.3.4	Multi-dwell protocol	5.2.1.2.1.4 Appendix J	102.1.3:M	yes__NA__
102.1.3.5	Frame synchronization subfield	5.2.1.2.1.5	102.1.3:M	yes__NA__
102.1.3.6	Message Indicator	5.2.1.2.1.6	102.1.3:M	yes__NA__
102.1.3.7	Transmission Word Count (TWC) subfield	5.2.1.2.1.7	102.1.3:M	yes__NA__

A.4.4 Synchronous mode transmission synchronization field

Item	Protocol Feature	Reference	Status	Support
102.2.3.1	Frame synchronization subfield	5.2.1.2.2.1	102.1.4:M	yes__NA__
102.2.3.2	Message Indicator	5.2.1.2.2.2	102.1.4:M	yes__NA__
102.2.3.3	Transmission word-count subfield	5.2.1.2.2.3 Appendix D	102.1.4:M	yes__NA__

A.4.5 Physical Layer (PL) to upper layer interactions

Item	Protocol Feature	Reference	Status	Support
102.4.1	PL-Unitdata request data/data length	5.2.4.a	M	yes
102.4.2	PL-Unitdata indication	5.2.4.b	M	yes
102.4.3	PL-status indication net activity	5.2.4.c	M	yes

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ANNEX B (informative)

CONCLUDING MATERIAL

B.1 Deviations from the Base Standards/Referenced Profiles

This MIL-STD documents the data contained in the MIL-STD-188-220A in the ISO/IEC TR 10000, Framework and Taxonomy of International Standardized Profiles and MIL-HDBK-829A format. This DSP does not deviate from the protocol as written in the base standard.

B.2 Subject Term (Keyword) Listing

CNR
Combat Net Radio
User Datagram Protocol
Internet Activities Board
Transmission Control protocol
Internet Protocol
Internet Message Control Protocol
ICMP
TCP
Network
Subnetwork
Physical Layer
HDLC

B.3 Preparing Activity:

DISA-JIEO (Project DCPS-0013, SP 02)

B.4 Reviewing Activity:

Army	SC, PT
Air Force	3, 17, 29, 33, 90
DLA	DH
DMA	MP
DIA	DI
DOT	OST
NSA	NS
OASD	IQ, DO, MA, IR
ODISC4	AC
NAVY	EC, CH, ND, TD, OM
USMC	MC, CG

B.5 Custodians:

DISA:	DC
Army:	SC
Air Force:	90
Navy:	OM
DIA:	DI
NSA:	NS
USMC:	MC
DLA:	DH
Other:	Joint Staff/Architecture & Integration
	USSPACECOM

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3 DOCUMENT TITLE Information Technology - DOD Standardized Profiles - Internet Transport Profile for DOD Communications - Part 6: Combat Net Radio (CNR)

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